MIL-F-37986
25 MARCH 1986

MILITARY SPECIFICATION

FORCEPS, HEMOSTATIC
Hartman, 4 Inch (10 cm)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers straight hemostatic forceps of the Hartman design.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and drawing. The following specifications, standards, and drawing form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

GG-I-526	-	Instruments, Dental and Surgical: General Specifications.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Set-up.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-L-10547 - Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)
24511 - FORCEPS, HEMOSTATIC, HARTMAN

(Copies of specifications, standards, and drawing required by contractors in connection with specification acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for.

E 92 - Vickers Hardness of Metallic Materials, Tests for.

E 140 - Hardness Conversion Tables for Metals (Relationship Between Brinel Hardness, Vickers Hardness, Rockwell Hardness, Rockwell Superficial Hardness and Knoop Hardness).

F 899 - Stainless Steel Billet, Bar, and Wire for Surgical Instruments

F 921 - Standard Deviation of Terms Relating to Hemostatic Forceps.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references coded herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The hemostatic forceps shall be in accordance with the applicable requirements of GG-I-526, and as specified herein.

3.2 Material. The forceps shall be fabricated from Class 4, in accordance with ASTM F899. All of the component parts of the instruments shall be made of martensitic stainless steel of Type 410, 410X, 416, 416 MOD, 420A and 420B.

3.3 Hardness. After proper stress relieving, Rockwell hardness shall be C40 to C48 when tested as specified in 4.4.1. The requirements of GG-I-526 shall apply.

3.4 Style, design and dimensions. The forceps shall be mosquito type of the Hartman design. The terminology of Hemostatic forceps shall be in accordance with ASTM F921. The forceps specified herein, shall be suitable for accomplishing hemostatis in delicate tissue. Both jaws of the forceps shall be serrated throughout their entire length. Serrations shall be consistent in depth and in accordance with GG-I-526. The forceps shall mesh completely and properly during closure. The action of the forceps shall be easy, without drag or binding action. The forceps shall be in strict accordance with Defense Personnel Support Center (DPSC) Drawing 24511 for the appropriate type and National Stock Number (NSN).

3.5 Locks. Shall be of box lock type. The box lock construction shall be accurately fitted, and shall be without burrs, or sharp edges at any place in the lock construction. Pin or rivet shall be permanently secured. The joint performance shall be smooth, of equal resistance, and non-binding in all positions when opening or closing the forceps (the lock may be lubricated). Further, under no circumstances shall looseness or free play of either shank permit any misalignment of the jaws in closing. NOTE: This paragraph shall replace the requirements specified in GG-I-526 for Locks.

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3.6 Finish. Finish shall be entirely mirror, satin or matte. Should the instrument be of mirror finish, the locks and ratchets shall be polished smooth, but need not possess a mirror finish. The matte finish shall exhibit a uniform nondirectional surface texture equivalent to that produced by lightly blasting the prepared surface with fine abrasives under pressure.

3.6.1 Passivation. The forceps shall be properly passivated.

3.7 Jaw and ratchet operation. When tested as specified in 4.4.2, the jaw tips shall close first and progressively tighten with each ratchet position. Each ratchet position shall hold firmly and release instantly.

3.8 Identification marking. In addition to the requirements specified in GG-I-526, forceps of foreign manufacture shall be marked "stainless".

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection shall be in accordance with GG-I-526.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the corrosion-resistant steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table I. The unit of product for examination purposes shall be one forceps.

TABLE I. Sampling criteria for inspection.

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in table II. The unit of product for test purposes shall be one forceps. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

TABLE II. Sampling criteria for tests.

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Forceps	Hardness test	3.3 & GG-I-526	4.4.1	S-2	1.5
	Dimensions	3.4	---	S-2	2.5
	Distortion test	GG-I-526	GG-I-526	S-2	2.5
	Copper sulfate test	GG-I-526	GG-I-526	S-2	1.0
	Boil test	GG-I-526	GG-I-526	S-2	1.0
	Jaw and ratchet operation test	3.7	4.4.2	S-2	1.0

4.2.3 Raw materials. A sufficient quantity of each lot of metal used in the manufacture of the forceps shall be selected for chemical analysis.4.3 Examination.

4.3.1 Examination of forceps. The forceps shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any forceps in the sample containing one or more defects shall be rejected and if the number of defective forceps in any sample exceeds the acceptance number for the sample, the lot represented by that sample shall be rejected.

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TABLE III. Classification of defects.

Categories	Defects
Critical	None defined.
Major	
101	Forceps not of the specified design for appropriate type, stock number and item identification.
102	Both jaws not serrated throughout full length.
103	Jaw serrations not of uniform depth and not well defined.
104	Jaw serrations not chamfered throughout full length.
105	Ratchet teeth not well defined.
106	Forceps not free from burrs, sharp edges, scale, rough areas or corrosion.
107	Finish of forceps not uniform and not as specified.
108	Serrations of both jaws do not completely mesh in fully closed position.
109	Box lock performance not as specified in 3.5.
110	Jaws possess sharp tips.
111	Ratchet teeth do not completely engage in all positions.
112	Pin for box lock not flush with surface and not free from burs.
Minor	
201	Forceps not free from dirt, polishing material or foreign matter.
202	Identification markings not present, not current, not complete, or not legible.

4.3.2 Packaging inspection. The inspection of the preservation-packaging shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Hardness test. Hardness shall be determined in accordance with ASTM Standards E 18 or E 92. Conversion of Vickers hardness values to Rockwell values shall be in accordance with ASTM Standard E 140.

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4.4.2 Jaw and operation test. The forceps shall be positioned to the first ratchet position, but not engaged. Forceps with jaw tips not touching shall be rejected; further, should all the jaw serrations touch, the forceps shall be rejected. The forceps shall then be engaged to each of the ratchet positions. Forceps that do not progressively tighten with each ratchet advance shall be rejected.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Each forceps shall be cleaned and dried with the applicable method as outlined in MIL-P-116.

5.1.1.2 Preservative. Preservative shall not be applied.

5.1.1.3 Unit packaging. Each forceps shall be packaged in accordance with MIL-P-116, method 1C-1.

5.1.1.5 Intermediate package. Twelve unit packages shall be packed in a box conforming to PPP-B-566, PPP-B-636, type CF, class domestic or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C. The level C preservation for forceps shall conform to MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B, and C as specified (see 6.2).

5.2.1 Level A. Thirty intermediate packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure, strapping, and weatherproofing shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Thirty intermediate packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for forceps shall conform to MIL-STD-794 requirements for this level.

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5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Levels A and B. Forceps, packed as specified in 5.2.1 and 5.2.2, shall be unitized on pallets conforming to MIL-STD-147.

5.2.4.2 Level C. Forceps, packed as specified in 5.2.3, shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B, and C. Each unit pack, intermedeiate package, exterior container, and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Intended use. The forceps shall be used during general surgical procedures.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. Applicable NSN.

6.3 National Stock Number coverage. This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6515-01-139-8672	Forceps, Hemostatic, Straight Mosquito, Hartman, 4 inch

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6.4 Matte and satin finish. It should be noted that matte finish specified herein can be distinguishable from mirror and satin under high-power illuminated magnification. Forceps of matte finish shall have a reduced reflective surface that is more glare resistant than satin finish.

MILITARY INTEREST:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITY

VA-OSS

Project No. 6515-3386

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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